SIMATS ENGINEERING

ASSESSMENT TOOL – 3

COURSE NAME: COMPUTER NETWORKDS

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COMPARITIVE ANALYSIS

1. **Star topology** is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.

2. **Mesh topology** offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.

3. **Bus topology** is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.

4. **Hybrid topology** is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.

5. **In terms of maintenance**, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

QUESTION 1)

Basis	***Star Topology***	***Bus Topology***
• **Structure**	All devices are connected to a central device such as a switch or hub.	All devices are connected to a single backbone cable.
• **Cabling**	Requires more cables because each device has a separate connection to the central device.	Requires less cabling because devices share one main cable.
• **Cost**	More expensive due to the central device and additional cables.	Less expensive because it requires fewer cables and no central device.
• **Reliability**	Highly reliable because failure of one device does not affect other devices.	Less reliable because failure of the backbone cable can bring down the entire network.
• **performance**	Provides better performance, especially when using switches.	Performance decreases as more devices are connected and traffic increases.
• **Installation**	Easier to troubleshoot and manage because each device has an individual connection.	Installation can be simpler initially, but identifying faults can be difficult.
• **Scalability**	Easy to add or remove devices without significantly affecting the network.	Adding more devices can increase traffic and reduce network performance.
• **Failure Impact**	Failure of the central switch/hub can affect the entire network.	Failure of the backbone cable can affect the entire network.
• **Data Transmission**	Data is usually transmitted through the central switch/hub to the destination device.	Data travels along the shared backbone, and devices check whether the data is intended for them.
• **Usage**	Commonly used in modern LANs, offices, schools, and businesses.	Mostly used in older or small networks because of its limitations.

QUESTION 2)

Basis	***Mesh Topology***	***Star Topology***
• **Structure**	Every node is connected to multiple or all other nodes.	All nodes are connected to a central switch or hub.
• **Fault Tolerance**	Very high fault tolerance because multiple paths are available.	Moderate fault tolerance because it depends on the central device.
• **Reliability**	Highly reliable because failure of one link usually does not affect communication.	Reliable for individual link failures, but central device failure can affect the entire network.
• **Cost**	Very expensive due to the large number of connections and cables required.	Less expensive than mesh because each device requires only one connection to the central device.
• **Cabling**	Requires a large amount of cabling, especially in a full mesh.	Requires less cabling because devices connect directly to the central device.
• **Performance**	Provides excellent performance because dedicated links reduce congestion.	Provides good performance, particularly when a switch is used, but traffic passes through the central device.
• **Scalability**	Difficult and expensive to expand because adding a node requires several new connections.	Easy to expand by connecting a new device to the central switch or hub.
• **Maintenance**	More difficult to maintain because of the large number of connections.	Easier to monitor, maintain, and troubleshoot.
• **Failure Impact**	Failure of one link or node generally does not stop the network because alternate paths are available.	Failure of the central switch or hub can bring down the entire network.

Basis	*Mesh Topology*	*Star Topology*
• **Applications**	Used in critical environments such as military, data centres, and backbone networks where reliability is important.	Commonly used in offices, schools, homes, and other LAN environments.

QUESTION 3)

Basis	Bus Topology	Mesh Topology
• **Structure**	All devices are connected to a single backbone cable.	Each node is connected to multiple or all other nodes.
• **Installation**	Simple and easy to install.	Complex and time-consuming to install.
• **Cabling**	Requires a small amount of cable.	Requires a large amount of cable.
• **Cost**	Low-cost because fewer cables and devices are required.	Very expensive because of numerous cables, ports, and connections.
• **Complexity**	Simple network structure with minimal connections.	Highly complex due to multiple connections between nodes.
• **Scalability**	Adding more devices can reduce network performance.	Adding devices requires many additional connections, making expansion difficult.
• **Reliability**	Less reliable because failure of the backbone can affect the entire network.	Highly reliable because alternate paths are available.
• **Fault Tolerance**	Low fault tolerance; a backbone failure can stop communication.	Very high fault tolerance; data can use alternative paths if a link fails.
• **Maintenance**	Easier to maintain because there are fewer connections.	More difficult to maintain because of the large number of connections.
• **Suitable For**	Suitable for small and simple networks where low cost is important.	Suitable for critical networks where high reliability and continuous communication are required.

QUESTION 4)

Basis	Hybrid Topology	Star Topology
• **Structure**	Combines two or more different topologies, such as star, bus, or mesh.	All devices are connected to a central switch or hub.
• **Scalability**	Highly scalable and can be expanded by adding new network segments.	Moderately scalable and depends on the capacity of the central device.
• **Flexibility**	Highly flexible because different topologies can be used in different sections.	Less flexible because all devices generally follow the same star structure.
• **Cost**	Generally, more expensive because it may require different networking devices and topologies.	Relatively less expensive because it usually requires one central device and individual cables.
• **Complexity**	More complex to design, configure, and manage.	Simpler to design, configure, and manage.
• **Expansion**	New segments can be added without significantly affecting existing sections.	Expansion is limited by the available ports on the central switch or hub.
• **Reliability**	Can provide high reliability depending on the topologies used in each section.	Failure of the central switch or hub can affect the entire network.
• **Performance**	Performance can be optimized by using different topologies for different network requirements.	Generally, provides good performance, especially when a switch is used.
• **Maintenance**	More difficult and costly to maintain because different network structures may be involved.	Easier to troubleshoot and maintain because of its simple structure.
• **Application**	Commonly used in large organizations, campuses, and enterprise networks.	Commonly used in homes, schools, offices, and small-to-medium LANs.

QUESTION 5)

Basis	Star Topology	Mesh Topology	Bus Topology	Hybrid Topology
• **Maintenance**	Easy to maintain.	Difficult to maintain.	Relatively difficult to maintain.	Moderately difficult to maintain.
• **Troubleshooting**	Faults can be easily isolated to a particular node or cable.	Troubleshooting is complex because of many connections.	Faults can be difficult to locate, especially in the backbone.	Troubleshooting depends on the different topologies used.
• **Fault Detection**	Easy to identify individual connection failures.	Can be time-consuming due to numerous links.	Difficult because a backbone fault may affect many devices.	Requires monitoring of multiple network sections.
• **Network Structure**	Simple and organized around a central switch or hub.	Complex, with multiple connections between nodes.	Simple structure using a shared backbone cable.	Complex because it combines different topologies.
• **Cable Management**	Relatively easy because each device has a separate cable.	Difficult because of the large number of cables.	Easy in terms of cable quantity, but backbone faults are difficult to handle.	Can be complicated because different cabling arrangements may be used.
• **Administration**	Requires basic networking knowledge.	Requires advanced networking skills.	Relatively simple to administer but fault identification can be difficult.	Requires skilled administration due to its mixed structure.

Basis	Star Topology	Mesh Topology	Bus Topology	Hybrid Topology
• **Repair**	Individual cables or devices can usually be repaired without affecting others.	Repairs can be complicated because multiple connections may be involved.	Backbone repairs can interrupt the entire network.	Repairs depend on the affected network segment.
• **Downtime**	Usually low because individual failures can be isolated.	Very low because alternative paths may be available.	Can be high if the backbone cable fails.	Depends on which topology or segment is affected.
• **Cost of Maintenance**	**Relatively low.**	**High due to numerous connections and equipment.**	**Low under normal conditions but backbone failures can be difficult to resolve.**	**Moderate to high due to its complexity.**
• **Overall Management**	**Easiest to manage.**	**Most difficult to manage.**	**Moderately difficult to manage.**	**Moderately complex to manage.**